

Predictive User Allocation for Stateful Edge Services

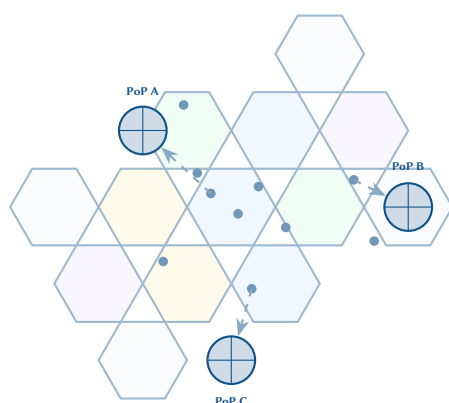
Stochastic Routing, Network Weather, and Dynamic Grid Indexing

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Abstract. This position paper explores a conceptual paradigm for geo-aware user allocation in stateful edge services—such as cloud gaming and remote desktops—driven by statistical learning rather than static routing. We argue that the *sticky allocation* constraint, which binds a user to a specific GPU resource for the duration of a session, necessitates a shift from reactive probing to predictive, stochastic modeling. By treating network topology as an evolving probabilistic map rather than a fixed graph, modern edge logic can aggregate large-scale telemetry—incorporating variables such as time-of-day, ISP prefix volatility, and session interactivity profiles—to build a predictive routing layer. This discussion formalizes the allocation problem as a constrained optimization and explores how Bayesian inference and background data preparation allow edge paradigms to continuously balance raw latency against economic efficiency within application-specific Quality of Experience (QoE) thresholds.

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1 The Allocation Problem for Stateful Edge Services

Latency-sensitive interactive services—cloud gaming, remote desktop, real-time collaboration, and live production tools—differ fundamentally from traditional Content Delivery Network (CDN) content delivery [0]. While a video stream or web page tolerates redirection mid-flow with minimal impact, a cloud gaming session carries GPU state, input buffers, and application context that cannot be cheaply migrated.

This *sticky allocation* constraint transforms the routing problem. In stateless systems, suboptimal placement is self-correcting. In stateful systems, a poor initial assignment degrades the entire session, typically lasting one to three hours. Network latency between user and assigned Point of Presence (PoP) dominates the Time to First Frame (TTFF), the critical optimisation target.

Insight:
Unlike HTTP, where any replica can serve any request, stateful sessions bind user and server for hours.

: Allocation Irreversibility

Once a user session binds to a PoP, the allocation remains immutable for the session lifetime. Allocation algorithms must therefore optimise for *expected* session-duration latency, not merely instantaneous connection-time latency.

Existing approaches to edge allocation fall into two broad categories. Domain Name System (DNS)-based systems [0] resolve user queries to the nearest edge using geo-DNS or latency-based DNS, but operate at resolver granularity rather than individual user granularity. Anycast-based systems [0] rely on Border Gateway Protocol (BGP) path selection, which optimises for AS-path length rather than latency [0]. Neither approach accounts for the sticky session constraint, the economic distinction between peered and transit paths, or the stochastic nature of user networking.

This position paper explores a conceptual paradigm driven by intelligent categorisation and large-scale statistical aggregation. We propose that modern edge controllers must shift from reactive probing to predictive routing:

- **Intelligent User Categorisation:** Distinguishing latency-critical (e.g., fast-paced gaming) from latency-tolerant (e.g., strategy games, remote desktop) sessions.
- **QoE-Bounded Cost Optimisation:** Routing tolerant sessions to more distant, cheaper facilities (e.g., via transit) provided the predicted RTT remains within the application’s maximum acceptable QoE boundary.
- **Periodic Background Preparation:** Batch-processing massive telemetry sets to refine time-of-day and ISP-prefix location models before real-time requests occur.

Insight:
Static routing rules fail against dynamic edge conditions. Aggregating telemetry from large user populations continuously trains stochastic models that capture routing topologies and congestion.

Feature	Legacy Approach	Proposed Paradigm
Routing Driver	Static IP/BGP Tables	Stochastic Probability Map
Spatial Resolution	City or ASN Level	Dynamic H3 Hierarchical Cells
Network Weather	Reactive (after failure)	Predictive (Bayesian Inference)
Metric Basis	AS-Path Length / Distance	Verified RTT Distribution
Cost Control	Manual / Policy-based	Automated QoE-Bounded Trade-offs
Multi-user Sync	Individual Best-Effort	Collective Center-of-Mass

Table 1. The Paradigm Shift: From static network assumptions to evolving stochastic models.

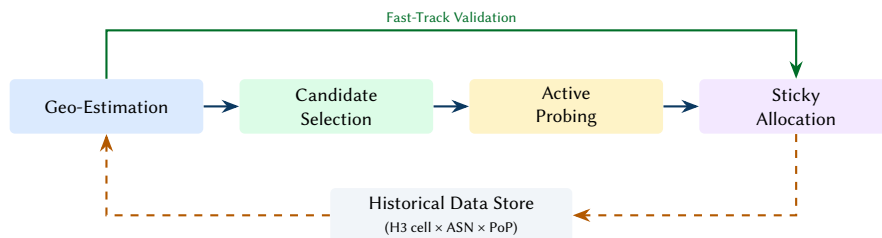


Figure 1. The allocation pipeline. Historical data creates a feedback loop that enables fast-track bypass when geo-estimation confidence is high.

2 System Model and Definitions

Stateful edge services must battle three adversarial constraints that render traditional web routing obsolete:

Insight:
Unlike stateless HTTP traffic, interactive edge sessions are deeply sticky. The initial allocation must therefore be correct for the entire duration of the session.

- **The Speed of Light:** Fiber optics impose a hard theoretical floor of $\approx 1\text{ms}$ of latency per 100km of distance. No amount of bandwidth can accelerate a packet once it is in transit.
- **The Chaos of Peering:** The internet is not a direct graph. A user physically located 50km from a datacenter may be routed through an exchange point 500km away due to arbitrary ISP peering disputes.
- **The Gravity of State:** Even though initial state transfers (e.g., loading a save game) are relatively lightweight, once a session starts, the user's live rendering context is locked to that specific GPU. "Hot-swapping" or live-migrating a session to another node if the network degrades is technically fraught and rarely viable at the edge.

To formalize these constraints, the domain maps to the tuple $\mathcal{S} = (\mathcal{P}, \mathcal{U}, \mathcal{H}, \mathcal{L})$, where:

- $\mathcal{P} = \{p_1, p_2, \dots, p_m\}$ is the set of PoPs where the service is deployed. In practice m is small ($m \approx 5-30$).
- $\mathcal{U}$ is the user population, identified by source IP prefix and Autonomous System Number (ASN).
- $\mathcal{H}$ is the H3 hexagonal tessellation of the service region at a chosen resolution.
- $\mathcal{L} : \mathcal{H} \times \mathcal{P} \rightarrow \mathbb{R}^+$ is the measured latency function, populated progressively through probing.

2.1 Illustrative Case Study: H3 Hexagonal Grids

Uber's H3 system [0] illustrates the conceptual approach. H3 provides a hierarchical tessellation of the globe into hexagonal cells, built on the icosahedral Discrete Global Grid System (DGGS) [0].

Hexagonal grids provide several advantages for stochastic allocation:

- **Uniform Adjacency:** Unlike rectangular grids, where diagonal neighbors are further away than orthogonal ones, all six neighbors of an H3 cell share an identical centroid distance. This uniformity reduces artifacts when interpolating latency across cell boundaries.
- **Hierarchical Containment:** H3 supports a nested hierarchy where larger cells perfectly contain smaller ones. This allows a predictive model to dynamically scale spatial resolution to match the variance of the underlying network block:

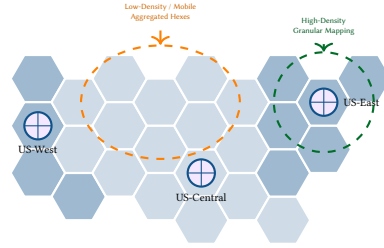


Figure 2. Continental H3 Mapping. A hierarchical grid can overlay the globe. Dense population centers (like the coastal edges) are subdivided into highly granular cells to map precise peering boundaries, while sparse or highly mobile regions are aggregated into larger cells to build statistical mass.

Predictive models dynamically scale spatial resolution to match the variance of the underlying network block:

- **Static Networks (High Resolution):** Residential ISPs and fixed-line networks can be mapped to highly granular geographic cells (e.g., area $\approx 3\text{ km}^2$). Because their physical origin rarely changes, routing logic can tightly bound the latency variance.
- **Mobile & Volatile Networks (Low Resolution):** Cellular prefixes (4G/5G) and VPN blocks exhibit massive physical variance. For these, a dynamic "zoom out" to a lower resolution (e.g., area up to $\approx 12,000\text{ km}^2$) can be applied. This expanded confidence bound aggregates the noisy telemetry, ensuring that routing logic achieves the necessary statistical mass before making a confident routing prediction.

Each H3 cell $h \in \mathcal{H}$ maintains a per-ASN, per-PoP latency record:

$$\mathcal{R}(h, a, p) = \{ \text{ewma}_{\text{rtt}}, \sigma_{\text{rtt}}, n_{\text{samples}}, t_{\text{last}} \}$$

where a is the origin ASN, p is the target PoP, and ewma_{rtt} is the Exponentially Weighted Average (EWA) of measured Round-Trip Times (RTTs).

Insight:
The H3 hex grid provides uniform-area cells with consistent adjacency—unlike rectangular grids which distort at high latitudes.

Insight:
Hexagons tile more effectively than squares for geographic analysis; they maintain a constant distance between a cell's center and all its neighbors. This property simplifies modeling network 'diffusion' across geographic boundaries.

Insight:
Spatial Coherence: Users within the same geographic cell and same origin ASN should theoretically observe latencies to a given PoP within a tightly bounded variance. If historical variance exceeds this bound, it suggests the cell is too large and must conceptually be subdivided into finer resolutions.

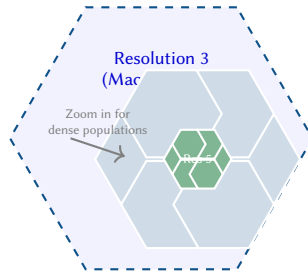


Figure 3. Hierarchical H3 Mapping. The H3 grid allows for seamless dynamic cell sizing. A large Resolution 3 cell covering a rural area or capturing mobile users can be recursively subdivided into 7 smaller Resolution 4 cells, which can be further subdivided into Resolution 5 cells for dense urban environments, allowing the statistical model to perfectly match geographic precision to user density.

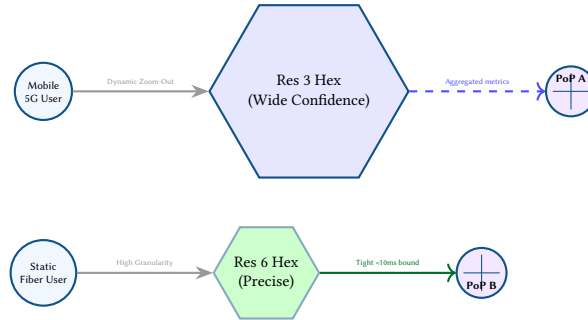
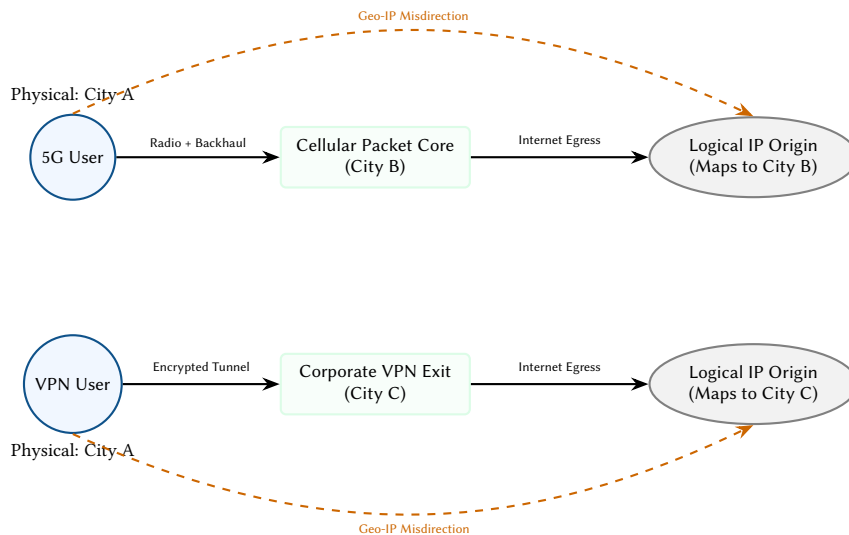


Figure 4. Conceptual Dynamic Cell Sizing. Instead of forcing all users into a uniform grid, predictive logic dynamically maps highly volatile connections (e.g., 5G, VPNs) to large, low-resolution hexes to capture their geographic variance, while static residential connections are mapped to precise, high-resolution hexes.

3 Geo-Estimation and Candidate Selection

Commercial Geo-IP databases routinely map 5G mobile users to the physical location of their cellular core network, not their handset—often misplacing them by hundreds of kilometers. Similarly, corporate VPNs obscure the origin entirely.



Insight:
To prevent mis-allocation, predictive frameworks refuse to trust a single source of truth. If signals conflict, they fall back to a wider probing strategy rather than making a blind assumption.

Figure 5. The Fragility of Geo-IP. Mobile 5G users are frequently routed through regional packet cores, while VPN users tunnel to distant corporate exits. In both cases, relying solely on the logical IP address results in massive geographic misdirection, routing the user based on the gateway’s location rather than their physical handset.

The first stage of the pipeline maps an incoming user connection to a geographic estimate and a short list of candidate PoPs. This stage must execute in under 10 ms to avoid adding perceptible delay to the connection setup.

3.1 Multi-Signal Geo-Estimation

IP geolocation databases are known to be imprecise, particularly for mobile networks and enterprise egress points [0]. Three independent signals combine to form the estimate:

1. **Bayesian-Enhanced Geo-IP Lookup:** Commercial databases (like MaxMind) provide a baseline (lat, lon) and confidence radius, but they are often stale or structurally inaccurate for mobile users. This baseline is continuously augmented through Bayesian inference. By integrating ground-truth telemetry from other co-located edge services—leveraging aggregate prefix latencies across the entire ecosystem—the framework dynamically expands or contracts the IP’s spatial confidence bound in real-time. If a broader platform service detects a sudden latency shift for a specific /24 prefix, that signal immediately informs the cloud gaming allocation, actively discounting the commercial Geo-IP data if it contradicts real-world measurement trends.

Insight:
No single Geolocation signals are rarely reliable in isolation. A robust framework must fuse multiple signals and degrade gracefully when they conflict.

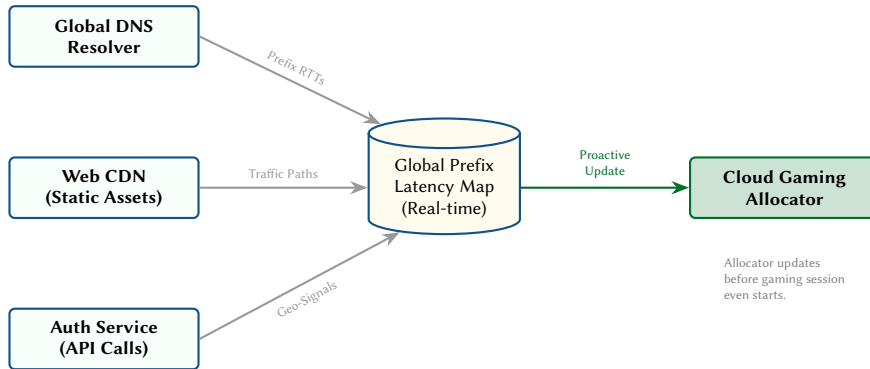


Figure 6. Cross-Service Telemetry Augmentation. Edge allocation does not operate in isolation. By ingesting real-time latency and path data from co-located services (DNS, CDN, Auth), predictive models maintain an always-fresh view of internet routing. This allows for detecting regional ISP shifts and updating predictive routing tables proactively.

2. **ASN-to-region mapping:** derived from Resource Public Key Infrastructure (RPKI) data and PeeringDB records [0], associating each ASN with a set of known service regions.
3. **Historical cell affinity:** if the user’s /24 prefix has been seen before, the previously assigned H3 cell provides a strong prior.

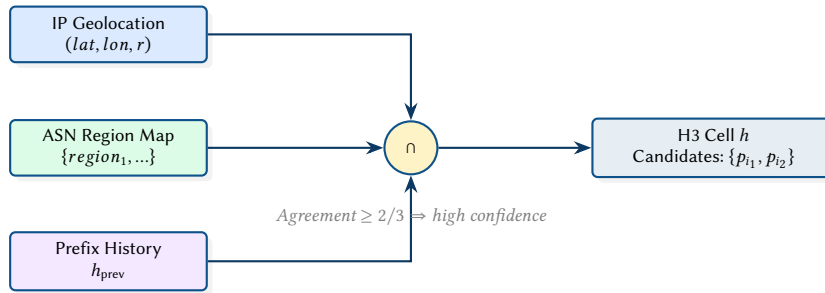


Figure 7. Multi-signal geo-estimation. The fusion gate requires at least two signals to agree on a region before assigning high confidence.

3.2 Dynamic Prefix Subdivision

While IPv4 /24 (or IPv6 /48) blocks are the industry standard for routing aggregation, treating a /24 as a single, homogenous geographic entity is often an oversimplification. If an ISP dynamically reallocates a /24 block across two different cities, or if a block is split between static residential users and a mobile NAT gateway, mapping the entire prefix to a single H3 cell will yield suboptimal allocations for half the users.

Dynamic Prefix Subdivision addresses this limitation. If the variance of latency measurements within a single /24 block exceeds a critical threshold, predictive models conceptually split the block into two /25 prefixes (or further into /26s) and track their telemetry independently.

This capability is drastically enhanced when the edge network is augmented with measurements from other products offered by the same provider. A cloud gaming service might only see a user for two hours a week, but if the same provider also runs a popular DNS resolver or web CDN, the aggregate cross-product telemetry can instantly identify when an ISP has geographically split a /24 block, allowing the gaming allocator to subdivide the prefix before the user even connects.

Insight:
A wide geographic footprint of data centers acts as a high-resolution lens. The more PoPs you have, the more easily you can detect when subsets of a single IP prefix are routing to different physical locations.

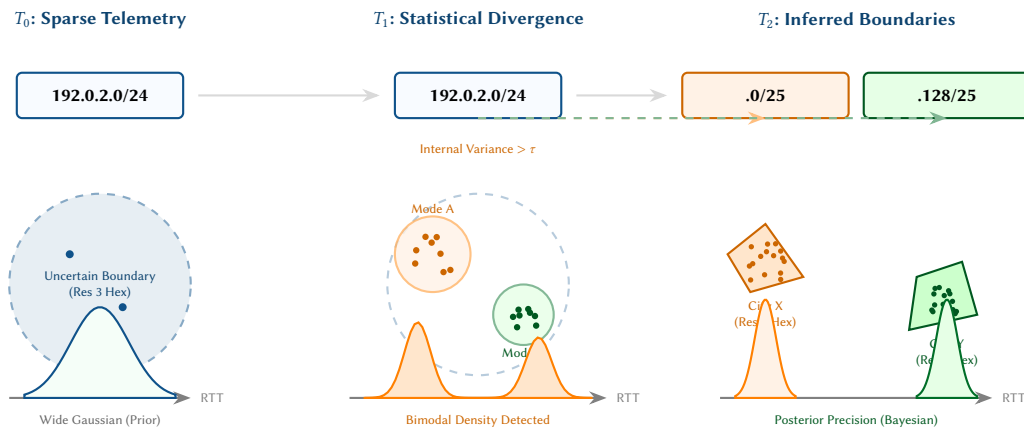


Figure 8. Temporal Prefix Subdivision and Spatial Inference. At T_0 , sparse latency samples map a standard /24 BGP block to a broad, highly uncertain geographic radius. At T_1 , as sample density increases, the stochastic model detects a bimodal distribution, revealing that the ISP has split the block. By T_2 , the model dynamically subdivides the prefix into /25s, abandoning the broad initial circle for tight, inferred geographic polygons (H3 micro-cells) based strictly on the separated latency profiles.

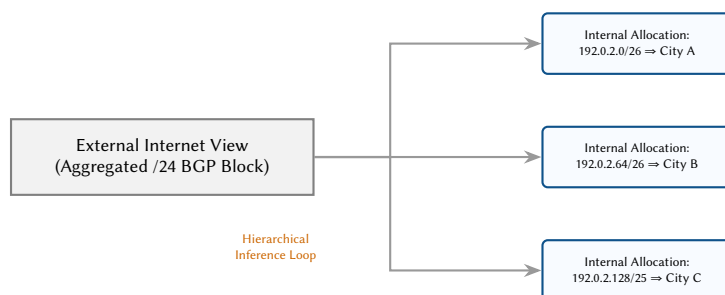


Figure 9. Micro-Granularity vs. BGP Abstraction. While the internet routing table sees a single aggregated /24 prefix, predictive paradigms maintain a micro-granular view of the block's internal topology. By subdividing the prefix internally based on latency signatures, sub-city routing precision is achieved without increasing the external BGP routing table size.

The geo-estimated coordinates are resolved to an H3 cell h at resolution 4. The candidate set is then:

$$C(h) = \text{top-}k(\mathcal{P}, d_{\text{haversine}}(h, p_i) + \alpha \cdot \text{cost}(h, p_i)), \quad k \leq 3$$

3.3 The IPv6 Sparsity Challenge

While IPv4 /24 blocks provide a strong prior, the transition to IPv6 introduces a profound statistical challenge: *address space sparsity*. Because IPv6 assigns massive /64 or /48 blocks to individual households, attempting to build a stochastic latency model per-prefix completely defeats the Law of Large Numbers. There are simply not enough users within a single /64 to ever build statistical mass.

To solve this, predictive models must employ **Aggressive Hierarchical Roll-Up** for IPv6 traffic. Rather than splitting prefixes as they do for IPv4, the model starts by aggressively aggregating IPv6 addresses up to the /32 or /29 level (often representing an entire ISP's regional allocation). If the variance within that massive block remains tight, the logic safely exploits it. Subdividing the IPv6 prefix (e.g., down to a /48) only occurs if bimodal latency divergence forces it, strictly conserving statistical mass.

Design Decision 1: Why $k \leq 3$ Candidates

Limiting the candidate set to at most three PoPs bounds the probing latency overhead. Each probe requires a round trip, so probing k candidates adds $k \times \text{RTT}_{\text{max}}$ to the connection setup. For a target TTFF of under 200 ms, $k = 2$ is typical; $k = 3$ is reserved for boundary cells where the nearest two PoPs have similar estimated latencies.

4 Active Probing and Fast-Track Validation

If geo-estimation provides a hypothesis, active probing provides the empirical proof. However, running a full battery of probes delays the Time-To-First-Frame (TTFF).

A **Fast-Track Gate** solves this. If a single lightweight probe returns an RTT that falls within the expected statistical variance ($\mu \pm 2\sigma$) of the pre-computed background model, the user is instantly allocated.

Geo-estimation provides a hypothesis; active probing validates it. The probing phase measures actual RTT from the user to each candidate PoP and compares it against historical expectations.

Insight:
Full multi-region evaluation is only triggered when the empirical probe explicitly violates the stochastic model's prediction.

4.1 The Probe Protocol

Upon connection, an allocation controller dispatches lightweight probes (ICMP echo or application-level TCP SYN/ACK) to each candidate PoP from a vantage point topologically close to the user's ingress.

Each probe returns a measured $\text{RTT}_{\text{obs}}(u, p_i)$. An allocation controller would then apply a validation gate:

Insight:
The probe must traverse the same network path the session traffic will use. Probing from a different vantage point measures a different path and produces misleading results.

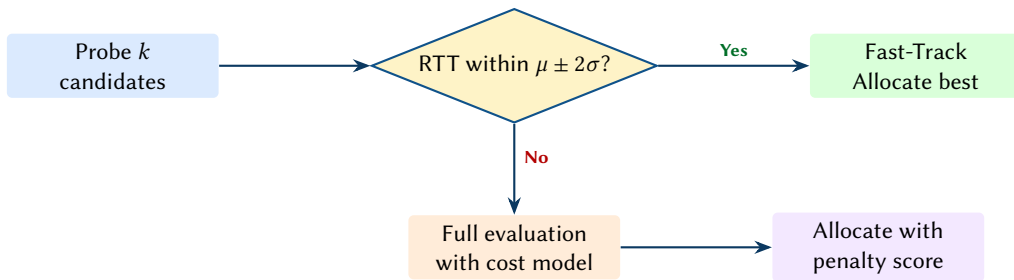


Figure 10. The validation gate. When probed RTT matches historical expectations, the user is fast-tracked to the lowest-latency candidate.

4.2 Fast-Track Criterion

A user is *fast-tracked* when the observed RTT to the best candidate falls within the expected range derived from historical data for that cell-ASN pair. Calculating the historical mean (μ) and standard deviation (σ) of the latency profile establishes a strict confidence bound.

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4.3 Preventing Overfitting in Sparse Cells

In geographic cells with sparse user populations, relying on a naive statistical average risks severe overfitting. A single anomalous probe could skew the distribution, creating a false "normal" that misguides future allocations.

Insight:
Fast-tracking is the steady-state path. Once a cell-ASN pair has accumulated ≥ 30 samples, over 90% of allocations should qualify for fast-track, reducing connection setup to a single probe round trip.

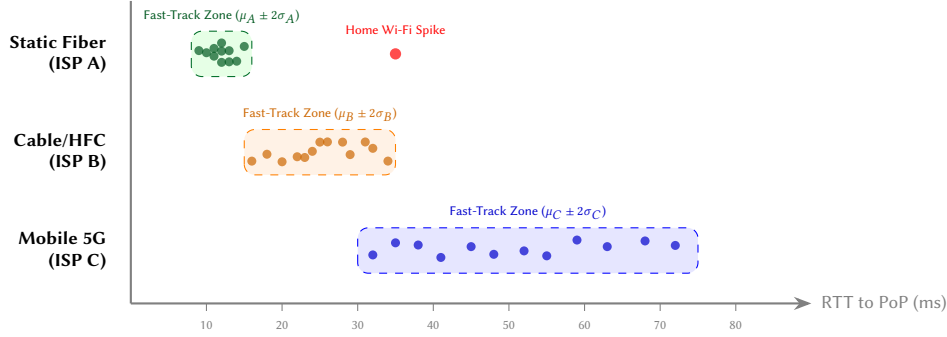


Figure 11. Disaggregated Latency Profiles. Even within a single geographic cell connecting to the same PoP, relying on a unified latency average is misleading. Different access technologies exhibit entirely different stochastic profiles. Intelligent allocation controllers calculate distinct confidence bounds ($\mu \pm 2\sigma$) for each sub-population (Cell $\times$ ASN), ensuring that a normal 50ms 5G measurement is fast-tracked, while a 50ms Fiber measurement is correctly flagged as a severe anomaly.

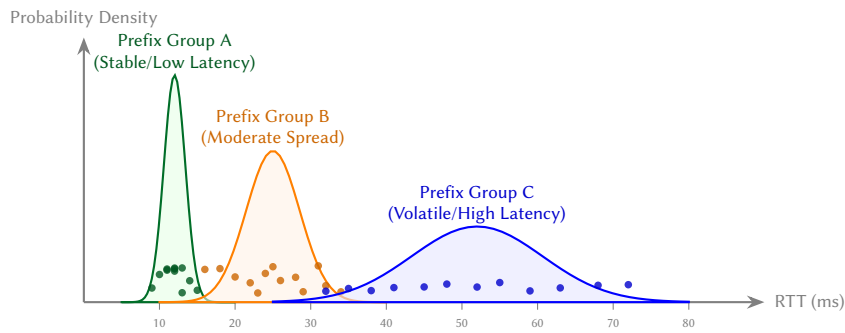


Figure 12. Continuous Statistical Distributions with Sample Overlay. Rather than treating an entire geographic cell as a single entity, the conceptual model identifies distinct performance capabilities (prefix groups) within it. As discrete samples (dots) accumulate, they converge into continuous probability distributions, allowing for an understanding that a 40ms RTT is a normal event for Group C, but a critical anomaly for Group A.

Bayesian inference mitigates this by dynamically adjusting the confidence bounds based on the sample density. When observations are limited, the model widens the acceptable variance, implicitly acknowledging its own uncertainty. As more telemetry accumulates, the Bayesian prior updates into a narrower posterior distribution, smoothly transitioning from a broad exploratory phase to a highly precise exploitation phase without relying on hard-coded sample thresholds.

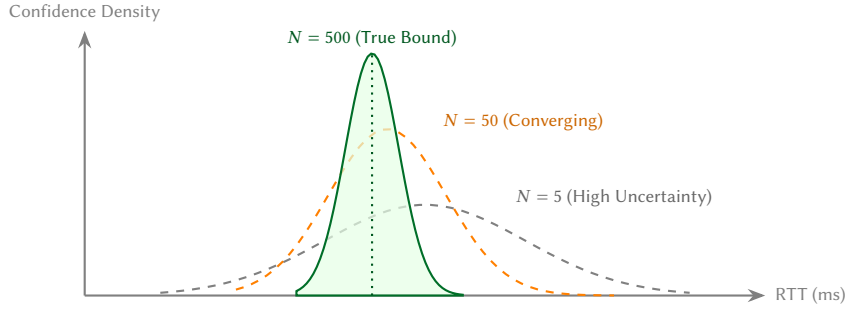


Figure 13. Statistical Convergence and the Law of Large Numbers. When a geographic cell has sparse telemetry ($N = 5$), the predictive model artificially expands the acceptable latency variance to prevent overfitting to early, potentially anomalous probes. As massive telemetry aggregates over time ($N = 500$), the Bayesian model dynamically tightens the confidence bounds, converging perfectly on the true underlying latency capability of that specific network segment.

: Probe Integrity

Probes must be indistinguishable from session setup traffic to ensure they traverse the same QoS queues, middleboxes, and interconnection points. A probe that bypasses a congested peering link produces an optimistic RTT estimate that will not reflect session-time performance.

5 Sticky Allocation and the Cost Model

Allocation decisions balance three competing objectives: latency minimisation, link-cost awareness, and capacity constraints.

5.1 Peering vs. Transit Economics

Interconnection type dictates the cost of delivering traffic from user to PoP [0]:

- **Settlement-free peering** at an Internet Exchange Point (IXP): zero marginal cost, though port saturation during peak hours limits capacity.
- **Paid peering / Private interconnect**: moderate cost with dedicated capacity; latency is typically lower than public IXP peering.
- **IP transit**: highest cost per Mbps; provides universal reachability and serves as the fallback when peering is unavailable.

Insight:
The cheapest path is not always the fastest. A transit link may offer lower latency than a congested peering exchange, but at higher cost.

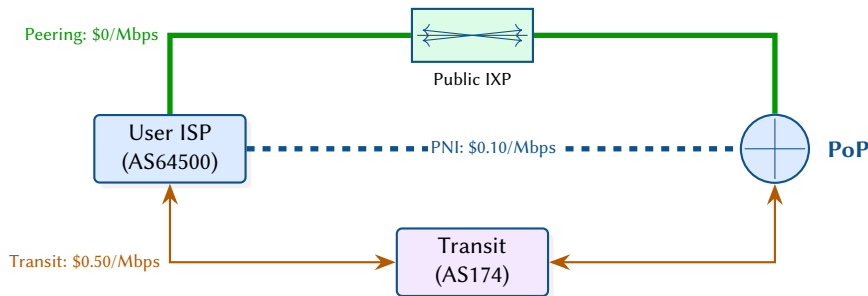


Figure 14. Three interconnection paths between a user's ISP and a service PoP, each with different cost and latency characteristics.

A cost function $\text{cost}(h, p)$ encodes these relationships, combining interconnection type with traffic volume:

$$\text{cost}(h, p) = \beta_{\text{type}}(a, p) \cdot \bar{B}_{\text{session}}$$

where $\beta_{\text{type}} \in \{0, \beta_{\text{pni}}, \beta_{\text{transit}}\}$ depends on the interconnection relationship between the user's ASN a and PoP p , and $\bar{B}_{\text{session}}$ is the expected bandwidth-duration product.

5.2 The Allocation Objective

Final allocation decisions consider a holistic score balancing three distinct dimensions:

1. **Raw Network Latency:** The stochastically predicted or empirically probed Time-To-First-Frame (TTFF).
2. **Economic Efficiency:** A cost penalty applied to transit routes relative to zero-cost peering.
3. **Capacity Load:** A penalty applied to edge nodes nearing maximum concurrent capacity.

A non-linear capacity penalty protects the user experience and maintains high utilisation. Edge nodes operate without penalty until they reach 70% capacity. Beyond this threshold, an exponentially steep penalty applies to the node's score.

Design Decision 2: Capacity Soft Limits

The quadratic penalty beyond 70% load acts as a soft limit that gradually redirects new sessions to alternative PoPs before hard capacity is reached. This avoids the cliff-edge behaviour of binary capacity gates, where a PoP at 99% load suddenly rejects all new sessions.

5.3 State Transfer and the Stickiness Constraint

While the baseline allocation objective optimises for raw network latency and capacity, achieving a true “instant connection” (TTFF) requires accounting for application-layer state transfer (e.g., retrieving a saved game from cold storage). Furthermore, mid-session network adaptation is highly restricted.

The immutable binding between session and PoP introduces a binary choice regarding state transfer:

- **Speculative Transfer:** Controllers pre-emptively replicate user save state to candidate PoPs based on offline stochastic models. This zeroes out setup penalty but increases background storage churn.
- **Allocation Penalty:** State transfer occurs on-demand during setup. The allocation equation (??) must include a setup-time penalty, $T_{\text{setup}}(u, p)$, representing the expected delay to fetch the user's save state to facility p .

Insight:
Once a user binds to an edge node, they are locked in. Live “hot-swapping” or migrating an active GPU rendering session to another facility to mitigate transient network degradation is technically complex and rarely yields a net benefit. Allocation must be optimal on the first attempt.

When enforcing strict TTFF targets, a suboptimal PoP that already caches the user's save state will frequently outscore a lower-latency facility requiring a remote fetch. Because migration is infeasible, the chosen PoP must sustain an acceptable QoE for the full session duration.

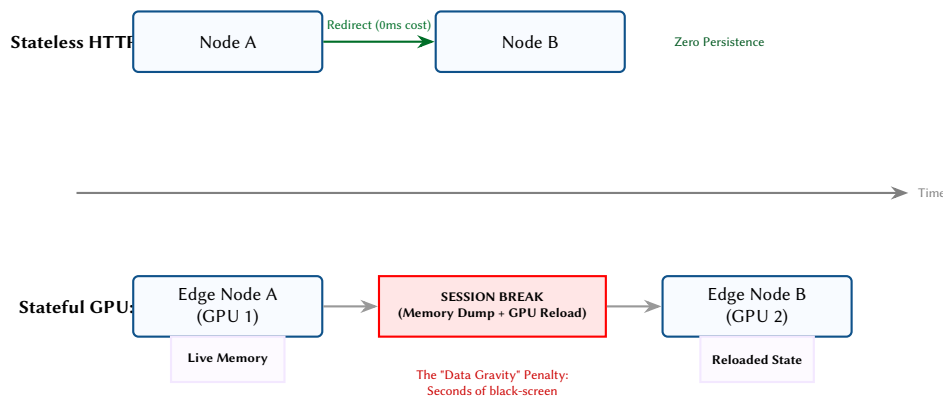


Figure 15. Data Gravity and the Handoff Penalty. Unlike stateless traffic which can be redirected at zero cost, stateful GPU sessions exhibit “data gravity.” Migrating an active session mid-stream requires a destructive memory dump and a full environment reload at the destination, resulting in seconds of session downtime. This penalty effectively renders the initial allocation decision permanent.

6 Regional Allocation and Network Weather

Physical proximity does not guarantee network proximity. Within the same metropolitan region, two users sitting in adjacent buildings may experience vastly different routing topologies based on their respective ISPs’ peering agreements and the prevailing “network weather” (transient congestion, route flaps, or localized fiber cuts).

The allocation logic continuously profiles these regional ISP differences. By relying on aggregate historical telemetry, it detects when a specific transit path becomes congested (e.g., peak evening hours) and proactively redirects affected users to a slightly more distant facility that maintains a stable, clean route, thereby preserving the Quality of Experience (QoE) despite localized network volatility.

Insight:
Users in the exact same city require different allocations if their ISPs peer at different exchanges. The stochastic model maps logical network topology, not just physical geography.

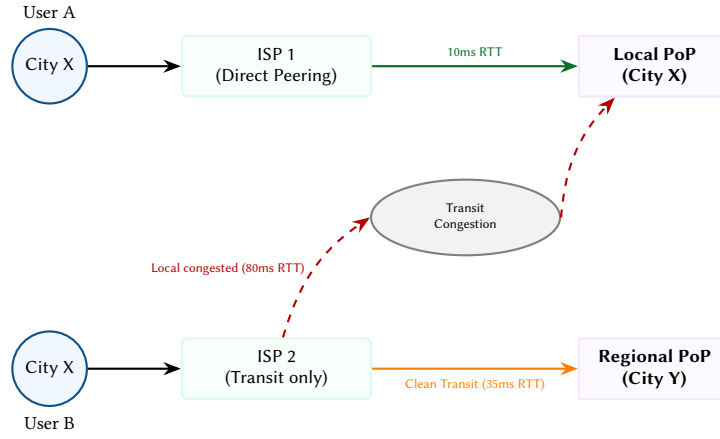


Figure 16. The impact of Network Weather. Despite sharing the same physical location (City X), User A is optimally routed to the Local PoP via direct peering, while User B’s ISP suffers transit congestion to the local facility. The stochastic model dynamically shifts User B to a further, but cleaner, Regional PoP in City Y.

6.1 Tie-Breaking and Determinism

When two candidates produce identical scores (within measurement noise $\epsilon = 1$ ms), the allocator applies a deterministic tie-breaker to prevent session flapping across PoPs:

$$\text{tiebreak}(p_i, p_j) = \text{hash}(\text{prefix}(u) \parallel \text{session_epoch}) \bmod 2$$

This ensures that users from the same prefix are consistently allocated to the same PoP within a given time epoch, improving cache locality and reducing cross-PoP state divergence.

: Stable Allocation

For any user u with unchanged network conditions, repeated allocation requests within the same epoch must yield the same PoP. This stability property prevents “PoP bouncing” between sessions and enables users to build affinity with nearby infrastructure.

6.2 Model Hysteresis and Stability

A core engineering challenge in stochastic allocation is preventing “PoP Flapping,” where a user is repeatedly reassigned between two candidate facilities with near-identical performance scores. Given the high cost of session initialization, constant switching due to minor measurement noise (< 2 ms) results in a net negative QoE.

To ensure stability, the framework can employ **Decision Hysteresis**. A user is only migrated to a new PoP if the predicted RTT improvement exceeds a stability threshold $\Delta\tau$. This creates a “dead zone” where minor network fluctuations do not trigger a disruptive shift in allocation, ensuring that session stickiness is preserved unless a significant, sustained performance gain is available.

7 Session Stickiness and Migration Constraints

The defining constraint of this class of services is that allocation is a *commitment* [0]. Unlike HTTP load balancing, where consistent hashing [0] distributes requests across a pool, stateful sessions carry substantial server-side state:

- **GPU context:** frame buffers, shader caches, render state (cloud gaming).
- **Application state:** in-memory document models, undo history, cursor positions (collaborative editing).
- **Input pipeline:** buffered keystrokes, mouse events, prediction state.
- **Encoder state:** video codec state, adaptive bitrate history, perceptual quality model.

Axiom 1 (Allocation Commitment). For a session s allocated to PoP p at time t_0 , the binding $s \mapsto p$ is immutable for $t \in [t_0, t_0 + T_s]$ where T_s is the session duration. Any reallocation requires full session restart with user-visible interruption.

Insight:
Stability over stateful edges. 25ms session initialization is a preferable trade-off to frequently chasing a fluctuating path.

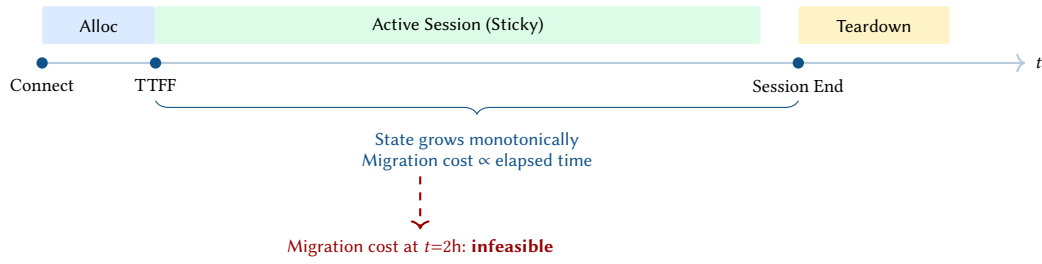


Figure 17. Session lifecycle. State accumulates throughout the active phase, making mid-session migration increasingly expensive until it becomes infeasible.

This commitment property has a direct implication for the allocation algorithm: the expected **RTT** over the *full session duration* matters more than the instantaneous **RTT** at connection time. A **PoP** that is marginally faster now but is approaching capacity (and will suffer congestion in 30 minutes) is a worse choice than a slightly slower **PoP** with headroom.

7.1 Handling Network Path Changes

Network paths are not static. **BGP** re-convergence, peering disputes, and traffic engineering changes can alter the latency between a user and their assigned **PoP** mid-session. While the allocation cannot change, a robust allocation strategy should:

1. **Detect** path changes via continuous in-band **RTT** monitoring.
2. **Record** the event in the historical data store, flagging the cell-ASN-PoP tuple as having elevated variance.
3. **Adapt** future allocations by widening the confidence interval σ for affected tuples, which forces new users through the full probe path rather than fast-track.

8 Learning from Accumulated Data

Every allocation produces a measurement that enriches the historical data store. Over time, this creates a dense latency map indexed by (h, a, p) —the H3 cell, origin **ASN**, and target **PoP**.

8.1 Exploration vs. Exploitation

The allocation problem exhibits a classic exploration–exploitation tension [0]. Exploiting known-good allocations maximises immediate user experience; exploring alternative **PoPs** discovers potentially better paths.

A contextual Multi-Armed Bandit (**MAB**) formulation operates where:

- **Arms** are the candidate **PoPs**.
- **Context** is the (h, a) tuple.
- **Reward** is the negative **RTT** (lower latency = higher reward).

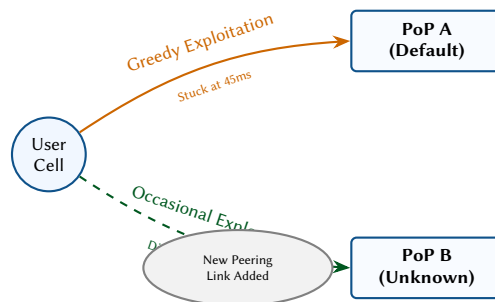


Figure 18. The Necessity of Exploration. A purely greedy routing model exploits the first acceptable route it finds (e.g., locking onto a 45ms path to PoP A) and never deviates. By intentionally allocating a small percentage of sessions to “under-explored” facilities, the topology is continuously mapped, discovering when underlying internet conditions change (e.g., finding a new 20ms path to PoP B).

The exploration bonus is applied only to cell-ASN pairs with sparse observations. Predictive models dynamically boost the priority of “under-explored” facilities, occasionally probing alternative paths to discover routing improvements rather than permanently locking onto the first acceptable **PoP**.

Insight:
The data store is the system’s primary asset. Raw latency measurements depreciate rapidly (hours), but the statistical aggregates they produce have value for weeks.

8.2 Temporal Decay and Staleness

Network conditions change. Historical measurements must be discounted over time to prevent stale data from dominating allocation decisions. Predictive models employ an exponentially weighted moving average, where recent telemetry aggressively overrides older samples.

Additionally, if no samples arrive for a specific network segment within 24 hours, the confidence bounds artificially widen. This forces allocation controllers to abandon the fast-track mechanism and issue fresh, empirical probes the next time a user connects from that region.

: Data Freshness

No allocation decision may rely on historical data older than 7 days without fresh probe confirmation. An allocation controller must treat stale cells as “unmeasured” and route through the full probe path rather than fast-track.

8.3 Intelligent User Categorisation and QoE-Aware Routing

While raw latency minimisation is the default objective, not all sessions exhibit the same sensitivity to network delay. Machine learning models classify user sessions, allowing allocation controllers to make nuanced, cost-aware decisions without violating Quality of Experience (QoE).

Modeling the user state combines historical session metadata and real-time Kalman filtering to predict session longevity and interaction intensity. Furthermore, the allocation logic natively integrates several business and application-layer constraints:

- **Game Heterogeneity:** Latency requirements vary drastically by genre. A simple platformer played casually can gracefully tolerate latency that would render a competitive first-person shooter unplayable. Intelligent controllers map application types to distinct maximum acceptable QoE bounds (RTT_{max}).
- **Premium vs. Standard Tiers:** Service-level agreements dictate routing priorities. “Premium” or “Pro” users are categorically exempt from cost-optimisation routing and are structurally guaranteed a best-effort local allocation (lowest absolute RTT), while standard-tier users may be routed to cheaper, more distant facilities provided they remain within the acceptable QoE threshold.
- **Multiplayer Synchronisation Constraints:** Multiplayer sessions inherently amplify latency sensitivity. In competitive or collaborative environments, the user experience relies not just on individual TTFF, but on state synchronisation with other players.

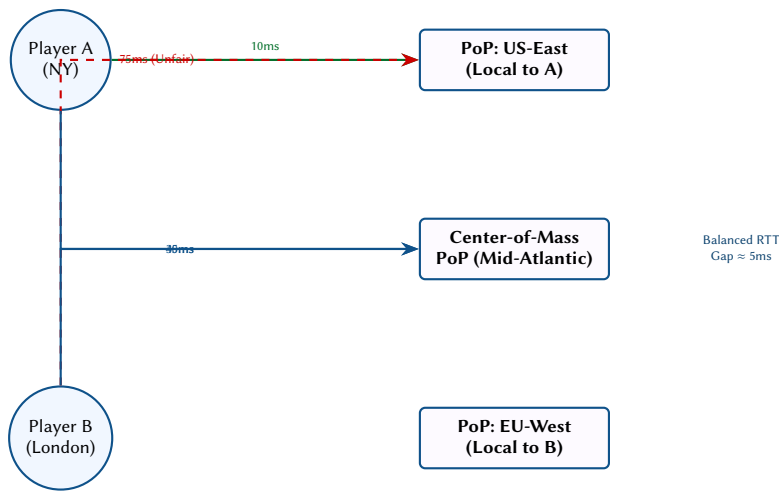


Figure 19. Multiplayer Center-of-Mass Allocation. In multi-user sessions, the objective shifts from individual latency minimisation to collective fairness. While routing to a local PoP (NY) is optimal for Player A, it creates a severe competitive disadvantage for Player B. A predictive model identifies a “Center-of-Mass” PoP that balances the latency gap between all participants, preserving synchronisation quality.

- **Kalman Filtering for Latency Tolerance:** By treating user input frequency as a noisy signal, a discrete Kalman filter estimates the underlying “interactivity state” of the session. If the predicted state falls below a QoE-critical threshold, routing controllers expand the acceptable latency radius.
- **Session Length Prediction:** Gradient boosted decision trees trained on user history and application type predict whether a session will be ephemeral (< 10 minutes) or persistent (> 2 hours).

When the model categorises a user as having high latency tolerance (e.g., standard tier, non-multiplayer, slow-paced game), allocation algorithms relax the fast-track criterion. Instead of assigning the absolute

minimum-RTT PoP, it selects the lowest-cost facility within the dynamically calculated acceptable QoE bound.

: QoE-Bounded Cost Optimisation

Cost-based routing must never override the application's maximum acceptable latency threshold. The allocation algorithm is only permitted to select a cheaper, more distant PoP if the estimated $RTT \leq RTT_{\max}$ for that specific user category.

8.4 Session Evaluation and Post-Mortem Analytics

Allocation is not fire-and-forget. A continuous evaluation is necessary to determine whether the predictive models resulted in a successful user outcome. Premature session termination is one of the strongest implicit signals of a failed allocation.

To prevent the model from prematurely penalising a clean network path due to normal user behavior, a **Post-Mortem Evaluation Loop** can be conceptualized as follows:

- **Expected Session Profiling:** Predictive models profile typical session lengths per game type and per user history.
- **Cross-Referencing Telemetry:** If a session terminates abnormally early, the loop cross-references the event with the live network telemetry. If the network was stable, the early termination is logged as behavioral and ignored by the routing model.
- **Penalty Assignment:** If the early termination correlates with high jitter, packet loss, or a sustained RTT spike, the allocation is flagged as "suboptimal." The underlying stochastic model aggressively penalizes that specific Cell-PoP path for future allocations, preventing other users from falling into the same degraded route.

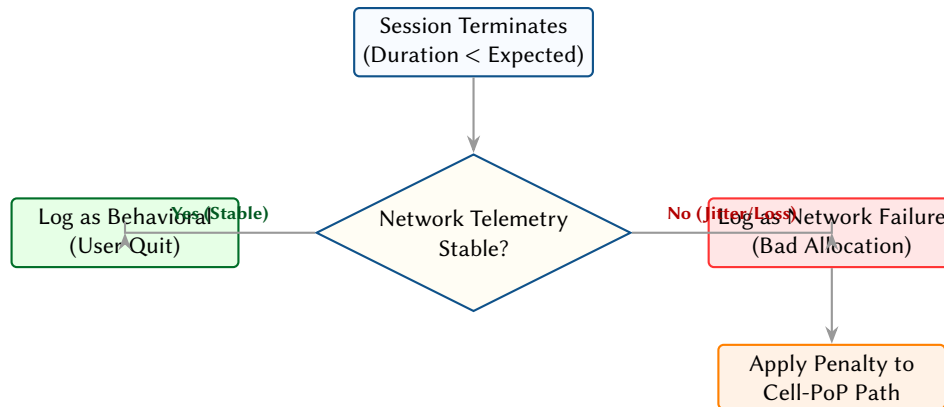


Figure 20. The Post-Mortem Evaluation Loop. To prevent the routing model from confusing normal behavioral churn (a user simply quitting a game early) with network failure, premature terminations are rigorously cross-referenced with live network telemetry. Only correlations with measured network instability trigger an algorithmic routing penalty.

8.5 Stochastic Model Building and Background Preparation

Real-time calculation of QoE bounds and predictive routing thresholds is computationally infeasible during a live session allocation. Instead, building a stochastic model of network conditions over time leverages the continuous telemetry stream from large user populations.

Continually feeding session outcomes back into the data store allows for training a probabilistic model that captures the underlying physics of the network, accounting for several non-deterministic factors:

- **Time-of-Day Congestion:** Routing performance follows cyclical patterns. A peering link providing 15ms latency at 04:00 may degrade to 80ms with high jitter at 20:00. The stochastic model incorporates local time-of-day as a primary feature, dynamically penalising paths prone to prime-time congestion.
- **ISP Prefix Anomalies:** A user's physical distance to a PoP does not guarantee a direct route. By correlating BGP AS-paths with the H3 spatial grid, routing models identify specific ISP prefix blocks that systematically route suboptimally (e.g., hairpinning traffic through another state).
- **Connection Volatility Profiles:** Different transport mediums exhibit distinct, identifiable latency signatures. A cloud gaming user playing on a 5G connection while driving will experience periodic spikes during cell-tower handoffs; similarly, satellite connections (e.g., Starlink) introduce distinct, semi-periodic latency jitter. Furthermore, users obscured behind VPNs present misleading IP geographic

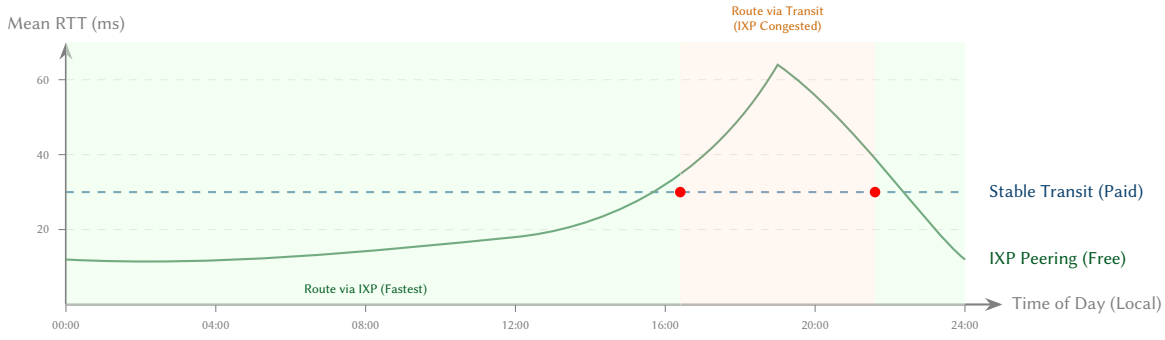


Figure 21. Cyclical Time-of-Day Congestion. Predictive logic tracks routing performance against the local time of the user’s timezone. Settlement-free IXP peering offers the lowest latency (< 15ms) for most of the day, but the port severely saturates during “prime time” evening hours. The framework anticipates this curve, dynamically shifting traffic to a more expensive, stable transit path precisely when the peering latency crosses the acceptable QoE threshold.

signals. By statistically profiling these signatures in the background, the allocation controller can apply specific mitigation strategies—such as routing high-volatility 5G users to a closer, albeit more expensive, edge node to buffer against jitter, or intentionally discounting anomalous RTT spikes to avoid erratic session migrations.

- **Stochastic Inference over Scale:** The law of large numbers allows for smoothing out transient noise. If a single user experiences a latency spike, it may be local Wi-Fi interference; if 500 users from the same AS block spike simultaneously, it is an infrastructure event. Predictive logic learns to classify and react to these aggregate anomalies in the background.

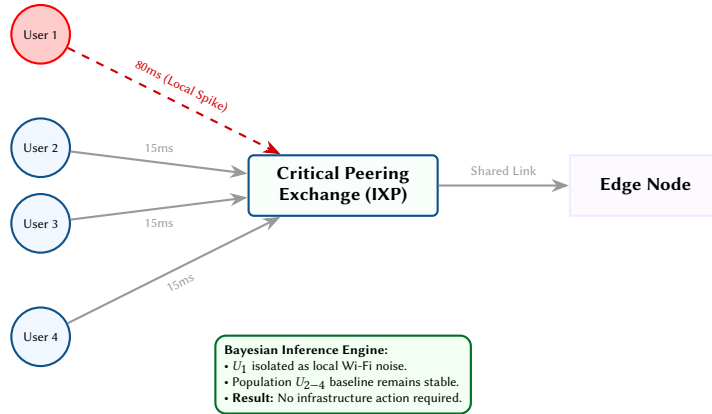


Figure 22. Stochastic Inference over Scale. By correlating telemetry from multiple users on the same logical path, the predictive model distinguishes between local “last-mile” noise (User 1) and systemic infrastructure failures. This prevents a single user’s hardware failure from polluting the aggregate routing tables for an entire region.

- **Outlier and Anomaly Isolation:** Network measurements inherently contain unpredictable edge cases—a user experiencing a 5000ms latency spike due to a failing home router, or a malfunctioning client reporting negative RTTs. Stochastic models employ robust statistical methods (e.g., median absolute deviation, isolation forests) during the batch-processing phase to identify and discard these extreme, non-representative outliers. This isolation ensures that transient, localized anomalies do not pollute the predictive routing tables and trigger unnecessary regional re-allocations for the broader user population.

This background preparation ensures that when a new session request arrives, allocation controllers perform an $O(1)$ lookup against the latest probabilistic model, retrieving a mathematically optimal decision for the user’s current context (time, location, and ISP).

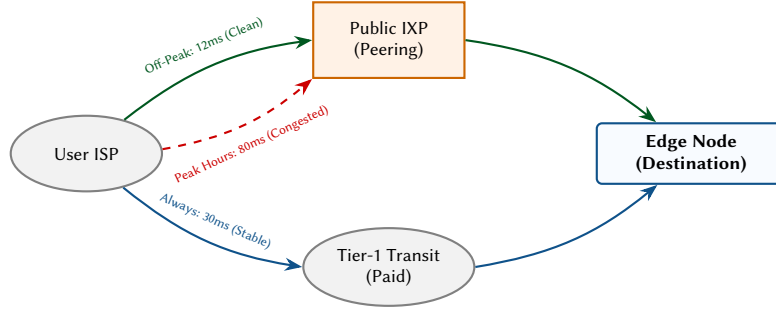


Figure 23. Time-of-Day Routing Economics. Settlement-free peering at a Public IXP is highly desirable during off-peak hours (12ms RTT, zero cost). However, during peak evening hours, the IXP port may saturate, causing extreme latency spikes and jitter. A predictive model anticipates this cyclical congestion and proactively reroutes latency-sensitive users over a slightly longer but highly stable Tier-1 transit link (30ms RTT, paid cost) to preserve QoE.

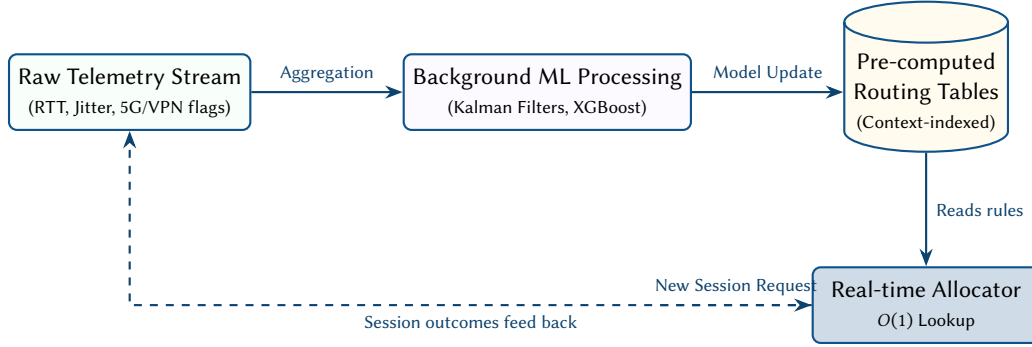


Figure 24. The Stochastic ML Pipeline. Heavy inferential tasks (Kalman filtering, connection volatility profiling) are shifted out of the critical connection path into periodic background batches, ensuring real-time allocations remain sub-10ms.

9 From Allocation to Capacity Planning

At a macro level, real-time allocation and long-term infrastructure investment represent two sides of the same generalized mathematical problem: the **Joint Allocation and Facility Location Problem (JAP-FLP)**. While allocation requires an $O(1)$ operational decision, capacity planning involves strategic optimization over months or years.

These two domains interact cyclically. Operational telemetry from individual allocations aggregates into a global demand map, informing infrastructure deployment to minimize system-wide costs (latency, deployment, and transit).

Insight:
Allocation data answers the question “where are users poorly served?” providing the necessary input for PoP placement and link augmentation.

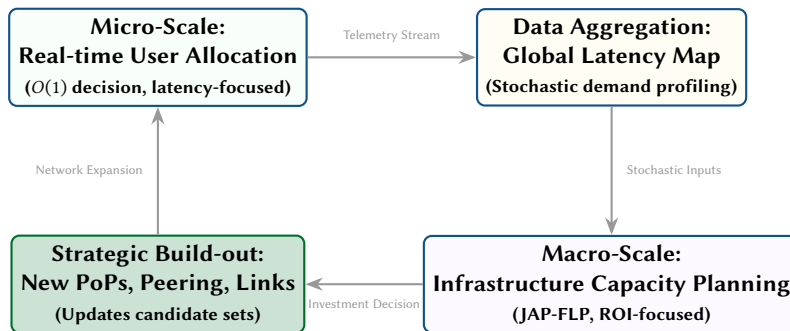


Figure 25. The Allocation-Capacity Macro Loop. Micro-scale telemetry from individual user sessions aggregates into a global stochastic demand profile. This profile serves as the primary input for macro-scale facility location problems. The resulting infrastructure build-out (new PoPs and peering links) directly updates the candidate sets available to the real-time allocator, completing the feedback loop.

9.1 Identifying Under-Served Regions

A cell h is *under-served* if the best achievable RTT to any PoP exceeds the Service-Level Agreement (SLA) threshold τ_{SLA} :

$$\text{under-served}(h) \iff \min_{p \in P} \mu(h, \cdot, p) > \tau_{\text{SLA}}$$

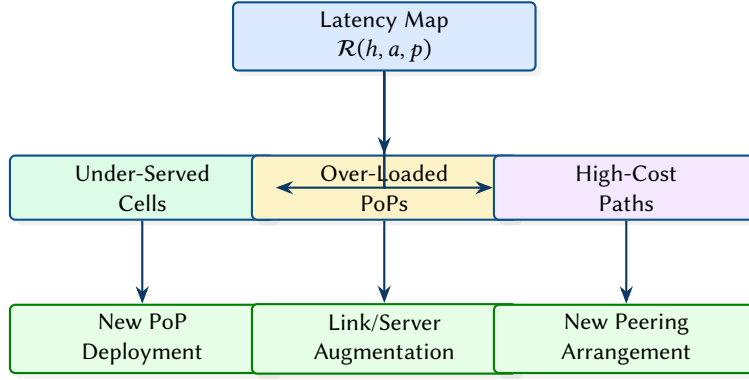


Figure 26. Capacity planning decision flow. Each category of deficiency maps to a distinct infrastructure action.

9.2 Peering Opportunity Detection

The cost model (section 5) reveals which **ASN-PoP** pairs carry significant traffic over transit links. When the traffic volume from a single **ASN** through transit exceeds a threshold V_{peer} , this flags a potential peering opportunity:

$$\text{peer_opportunity}(a, p) \iff \beta_{\text{type}}(a, p) = \beta_{\text{transit}} \wedge V(a, p) > V_{\text{peer}}$$

This creates a direct, data-driven input to the peering strategy [0], quantifying the cost savings of establishing a direct interconnect at the relevant **IXP** or via private network interconnect.

9.3 PoP Placement Optimisation

For longer-term planning, the H3 cell demand map enables a facility location analysis. The objective is to find the placement of m' new **PoPs** (where $m' \ll |H|$) that minimises the population-weighted maximum latency:

$$\mathcal{P}_{\text{new}} = \arg \min_{\mathcal{P}' \subseteq H, |\mathcal{P}'|=m'} \sum_{h \in H} w(h) \cdot \min_{p \in \mathcal{P} \cup \mathcal{P}'} d(h, p)$$

where $w(h)$ is the user-session demand weight for cell h , derived from allocation history. This is a weighted k -median problem, solvable via standard facility location heuristics [0].

Design Decision 3: Feedback Loop Integrity

The capacity planning system must consume the same data store that drives real-time allocation. Maintaining a separate “planning database” risks drift between operational reality and planning assumptions. One data store, two consumers.

10 Operational Considerations

10.1 Cold Start

When a new **PoP** is deployed or a new geographic region is served, no historical data exists. A predictive model must bootstrap intelligently:

1. Use geo-distance as the sole ranking signal for the first $N_{\text{bootstrap}}$ allocations per cell.
2. Set exploration rate c to its maximum value, ensuring all candidates are probed uniformly.
3. After $N_{\text{bootstrap}} = 50$ samples per cell-ASN pair, transition to the standard Upper Confidence Bound (**UCB**) regime.

10.2 Anomaly Detection

Sudden latency spikes (e.g., from a fibre cut or routing leak) must be detected and handled without waiting for the slow EWMA to adapt. A robust evaluation loop monitors for:

$$\text{anomaly}(h, a, p) \iff \text{RTT}_{\text{obs}} > \mu + 4\sigma \vee \text{RTT}_{\text{obs}} > \tau_{\text{SLA}}$$

When an anomaly is detected, the affected cell-ASN-PoP tuple is immediately marked as “suspect” and all new allocations for that tuple bypass fast-track until $N_{\text{clear}} = 10$ consecutive normal observations restore confidence.

10.3 Adversarial Routing and Latency Spoofing

When allocations dictate user experience, users will attempt to manipulate the process. Competitive players may attempt to force allocation to a specific region (e.g., to join overseas matches or bypass skill-brackets) by using VPNs or artificially delaying connection probes.

Robust frameworks defend against latency spoofing via **Server-Side Passive Validation**. While initial allocation relies on client-side probes, the edge node passively monitors actual session traffic (e.g., via TCP sequence acknowledgements). If the passive RTT remains consistently low (e.g., 20ms) despite an initial high probe report (e.g., 170ms), the framework identifies spoofing. Discarding the anomalous probe from background stochastic models preserves data integrity, and restricted fast-track privileges may apply to the offending account.

Insight:
A client artificially delaying probes by 150ms handicaps its own session. However, the stochastic model must exclude this spoofed data to avoid falsely penalising local PoPs for legitimate users.

10.4 Privacy and Data Minimisation

The data store indexes by H3 cell and [ASN](#), not by individual user IP. Prefix-level data (/24 for IPv4, /48 for IPv6) is retained only for the fast-track lookup and is purged after 7 days of inactivity.

11 Limitations of the State of the Art

Traditional network routing and content delivery models were designed for stateless HTTP traffic, where requests can be infinitely redirected. Applying these legacy architectures to stateful, latency-critical edge workloads reveals severe structural limitations.

Insight:
H3 resolution 4 cells contain tens of thousands of users, providing natural k-anonymity. Individual user behaviour is never tracked—only aggregate network path characteristics.

BGP Anycast is frequently presented as the default solution for edge routing. However, Anycast routes based on AS-path length, not network latency. Because it is unaware of peering congestion, an Anycast path may inadvertently route a user through an overloaded public exchange simply because it requires fewer hops.

CDN Request Routing. Akamai's mapping system [0] and its successor [0] pioneered end-user mapping for content delivery. Modern conceptual discussions must extend this to stateful sessions where the allocation is irreversible. While standard CDNs rely on massive server footprints where cached static content is ubiquitous, interactive edge workloads (like cloud gaming) rely on expensive, power-hungry GPUs. This results in an inherently sparse physical footprint compared to a traditional CDN. This sparsity, combined with tight latency bounds ($< 20\text{ms}$), renders naive Anycast or simple Geo-DNS unviable; a missed allocation to a distant GPU PoP results in an unplayable session, whereas a missed CDN allocation merely results in slightly higher content fetch times.

Insight:
Anycast lacks connection stickiness; a BGP route flap mid-session will sever a stateful connection instantly.

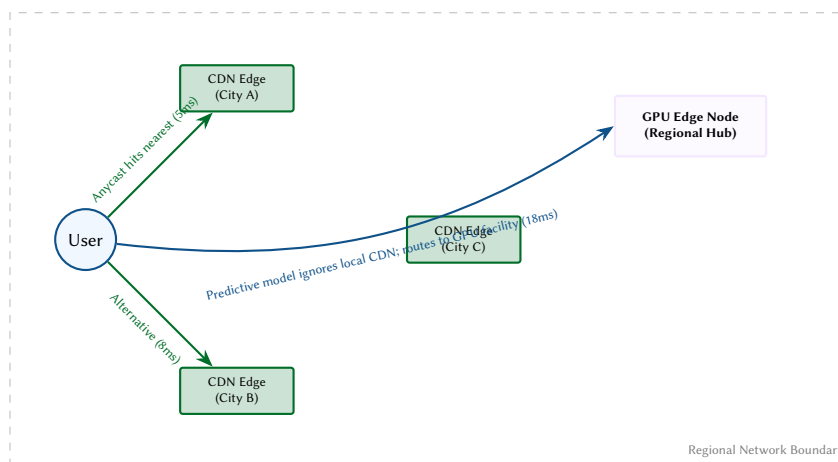


Figure 27. Infrastructure Sparsity and the Failure of Proximity. Standard CDNs enjoy a high-density deployment where static content is ubiquitous. Interactive GPU workloads are constrained by a sparse, regional footprint. Relying on Anycast to hit the "nearest" edge frequently results in a routing "miss" where the traffic terminates at a local node lacking the necessary compute resources.

Anycast Performance. Li et al. [0] and Calder et al. [0] characterise anycast CDN performance, showing that BGP-based anycast frequently selects suboptimal servers. Probing-based validation addresses this by measuring actual RTT rather than relying on BGP path selection.

Cloud Gaming Latency. Claypool and Claypool [0] establish latency thresholds for interactive gaming. Shea et al. [0] survey cloud gaming architectures. Lee et al. [0] introduce speculative execution to mask latency. A complementary approach is to optimise the network path before speculation is needed.

Load Balancing. Maglev [0] and Ananta [0] provide consistent hashing for connection-level stickiness at scale. Stateful edge routing builds on their session-affinity guarantees but adds latency-awareness to the initial placement decision.

Traffic Engineering and Steering. B4 [0] and SWAN [0] optimise WAN traffic allocation with centralised controllers. However, Traffic Engineering (TE) and backbone traffic steering (like MPLS or Segment Routing) are fundamentally insufficient for inbound edge routing. TE controls how traffic moves *inside* a provider's backbone or where it egresses. It cannot steer inbound user traffic coming across the public internet before it hits the ingress edge. Because a stateful cloud gaming session must bind to a specific GPU immediately upon connection, the "first mile" routing from the user's home ISP to the edge facility is entirely out of the provider's TE control. Therefore, the initial PoP allocation decision is the only effective lever to dictate the end-to-end network path.

Active Measurement. RIPE Atlas [0] provides a distributed measurement platform. Paris Traceroute [0] addresses measurement accuracy. The described probing system is simpler—single-hop RTT measurement—but must operate inline with connection setup, imposing strict latency budgets on the measurement itself.

12 Conclusion

Stateful edge services demand a fundamentally different allocation strategy from stateless content delivery. The irreversibility of session binding means that the allocation decision must be near-optimal at connection time, informed by geographic estimation, validated by active probing, and refined by accumulated historical data.

The conceptual paradigms discussed in this paper—geo-estimation with dynamic H3 spatial indexing, Bayesian multi-signal fusion, fast-track validation, and contextual bandit-based exploration—provide a principled framework for addressing the unique constraints of stateful edge workloads. By indexing latency data at the (h, a, p) granularity, predictive models can converge to highly accurate allocations while directly informing infrastructure investment decisions about PoP placement and peering strategy.

The core argument is that real-time allocation and long-term capacity planning are not separate problems, but two perspectives on the same underlying stochastic map. This integrated approach offers a pathway to achieving the near-instant connectivity required by the next generation of interactive edge services.

12.1 Future Research Frontiers

The conceptual paradigms established in this paper open several avenues for further theoretical and empirical exploration:

- **Cross-ISP Collaborative Telemetry:** Investigating the privacy-preserving mechanisms required for ISPs to share anonymous, aggregate routing health data directly with edge providers to eliminate the "measurement lag" in current stochastic models.
- **Federated Edge-Learning Paradigms:** Exploring how Bayesian model updates can be distributed across the PoP footprint, allowing for hyper-local "network weather" detection without centralizing massive telemetry streams.
- **Quantum-Safe Latency Probing:** As network security evolves, the impact of post-quantum cryptographic connections on initial connection handshakes and TTFF will require new predictive masking techniques.

Glossary of Terms

ASN	Autonomous System Number	3
BGP	Border Gateway Protocol	2
CDN	Content Delivery Network	2
DGGS	Discrete Global Grid System	3
DNS	Domain Name System	2
EWA	Exponentially Weighted Average	3
IXP	Internet Exchange Point	9
MAB	Multi-Armed Bandit	12
PoP	Point of Presence	2
RPKI	Resource Public Key Infrastructure	5
RTT	Round-Trip Time	3
SLA	Service-Level Agreement	16
TTFF	Time to First Frame	2
UCB	Upper Confidence Bound	17

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